QUANSER
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Mechatronic Sensors Trainer

Fundamental Labs – Color Sensor

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Color Sensor Lab

Why Explore Color Sensor?

Color sensors enable practical applications like color-accurate LED control, object recognition, display calibration, and ambient-light–aware systems. By understanding how the sensor perceives and quantifies color, we can design devices that respond correctly to real-world lighting, with great implications for robotics, smart lighting, quality inspection, and human-machine interfaces.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**021_color_sensor**

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of **color sensors**. The lab will have you explore the raw data from the sensor and be familiar with the process of calibration to convert sensor values to useful color values. Finally, you will compare the estimated color from the sensor signal to the ground truth color on your monitor.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.